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Changes in maternal morbidity and infant outcomes following state-level abortion bans post-Dobbs: a comparative interrupted time series study

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Abstract

Background The Supreme Court's decision in *Dobbs v. Jackson Women's Health Organization* overturned the *Roe v. Wade* precedent, significantly altering abortion access across the United States (U.S.). This ruling enabled states to regulate, limit, or ban abortion, resulting in millions of women losing access to abortion services and potentially affecting birth outcomes. This study aimed to assess the impact of the *Dobbs* decision on birth outcomes in states with and without abortion bans.

Methods A retrospective cohort analysis applied comparative interrupted time series to assess the difference in changes in adverse birth outcomes pre-and post-*Dobbs* across abortion legislation status. Natality data from the CDC WONDER database was analyzed. States were classified based on the presence or absence of abortion bans as of June 24, 2022. Proportions of non-living births, births with congenital anomalies, and maternal morbidities among U.S. births between January 2021 and February 2024 were examined.

Results No statistically significant difference in non-living births was observed between states with and without abortion bans. In states with abortion bans, the rate of births with congenital anomalies increased slightly post-*Dobbs* but the increase was not statistically significant. States without bans saw a decrease of 2 per 10,000 births ($p < 0.001$) from the baseline. The difference in the change in congenital anomalies between states with and without bans was 4.3 per 10,000 births ($p = 0.015$). States with abortion bans did not experience significant changes in maternal morbidity rates post-*Dobbs*, while states without bans experienced an increase of 4.8 per 10,000 births ($p < 0.001$). The difference between the changes in maternal morbidity rates was 4.5 per 10,000 births ($p = 0.014$).

Conclusions The *Dobbs* decision has led to divergent birth outcomes in states with and without abortion bans. States without bans experienced a decrease in congenital anomalies and an increase in maternal morbidity rates, while the proportion of these adverse birth outcomes were unchanging in states with bans. Although a direct evidence of abortion bans increasing adverse birth outcomes was not found, the difference in the changes in congenital anomalies across abortion policies suggests that there is a potential for the exacerbation of inequities and further investigation is warranted.

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Keywords Dobbs v. Jackson, Dobbs decision, Abortion bans, Maternal morbidity, Non-living birth, Congenital anomaly, Comparative interrupted time series

Background

The Supreme Court decision in *Dobbs v. Jackson Women's Health Organization* (Dobbs decision) overturned the 50-year precedent set by *Roe v. Wade*, resulting in a significant shift in abortions access across the United States (U.S.). This ruling removed federal protection for abortion, allowing states to regulate, limit, or ban the procedure. The Dobbs decision has paved the way for 28 states to implement laws that nearly entirely ban abortion through new legislation or pre-existing trigger laws previously unenforceable [1], leading to approximately 33 million U.S. women losing access to abortion services [2].

Prior studies have highlighted the critical importance of safe access to abortions in preventing maternal deaths, with estimates suggesting that up to 13% of maternal deaths could be prevented with safe abortion access [3]. Moreover, restrictions on abortion have been associated with increased infant mortality [4, 5]. While the regulation and restriction of abortion services are often intended to protect fetal health, they have broader public health implications, potentially forcing individuals to resort to unsafe abortion methods that fail to meet healthcare standards [6]. This is particularly concerning, given that many states with abortion bans already have poor maternal and infant mortality rates and limited access to care [7].

Furthermore, universal abortion bans are estimated to increase the incidence of congenital cardiac defects, associated morbidity, and resource utilization [8]. Infants born with congenital anomalies have special medical needs, and parents often face financial, emotional, logistical, physical and mental health, and parenting challenges [9–11]. The Dobbs decision removed the protection of the decision to terminate a pregnancy likely to lead to a lethal congenital anomaly, thereby reducing the burden on the parents and potentially alleviating the suffering of the child. The impact of the Dobbs decision must be assessed to understand the support needed for children with congenital anomalies and their families, who may feel unequipped to care for a child with additional and heavy medical needs, given their physical, mental, or financial capabilities.

The disparity in maternal care is stark, with 39% of counties in states banning abortion being considered maternity care deserts, compared to 25% of counties in states without abortion bans [7]. Given that most reproductive-age people in the U.S. now live in states banning abortion [12] and considering the increased potential

for significant adverse health outcomes, it is crucial to assess the impact of the Dobbs decision on maternal and infant outcomes in states that did and did not implement abortion bans. Previous studies have assessed the role of these bans on individual states [13], on fertility [14], and infant outcomes by race and ethnicity [5]. This study expands the existing literature by using a retrospective cohort analysis of nationwide natality data to evaluate the changes in maternal and infant birth outcomes following the Dobbs decision. By employing a comparative interrupted time-series (CITS) analysis design [15, 16], this study aims to provide an assessment of the association of the Dobbs v. Jackson ruling on public health outcomes related to abortion legislation, addressing gaps in prior research and offering new insights into the consequences of restricted abortion access.

Methods

A retrospective review of birth data within the U.S. was conducted to investigate the association between the Dobbs v. Jackson Women's Health Organization ruling and changes in maternal and infant birth outcomes in states with and without abortion bans. A CITS was used to evaluate differences in outcome rate changes before and after the Dobbs decision, comparing states with and without abortion bans.

Data source and study population

This retrospective cohort study used data from the Centers for Disease Control and Prevention's (CDC) WONDER (Wide-ranging Online Data for Epidemiologic Research) database, a publicly available database with information on live births by state of maternal residence [17]. The study cohort included all births reported to the CDC WONDER data in the United States from January 2021 to February 2024. All observations were included, and no exclusion criteria was applied. The final sample included 3,032,018 births in states with abortion bans and 5,495,792 births in states without them.

Classification of states

States were classified based on the presence or absence of abortion bans as of June 24, 2022 [18]. An abortion ban is defined as a state-imposed prohibition on abortion at any gestational age, including total bans and gestational limits (e.g., 6-week or 12-week bans). The classification included states with bans (Alabama, Arkansas, Idaho, Indiana, Kentucky, Louisiana, Mississippi, Missouri,

North Dakota, Oklahoma, South Dakota, Tennessee, Texas, and West Virginia) and states where abortion is legal beyond 22 weeks (California, Colorado, Connecticut, Delaware, the District of Columbia, Hawaii, Illinois, Maine, Maryland, Massachusetts, Michigan, Minnesota, Montana, Nevada, New Hampshire, New Jersey, New Mexico, New York, Oregon, Pennsylvania, Rhode Island, Vermont, Virginia, Washington, and Wyoming).

Outcomes of interest

The primary outcomes of interest included non-living births, births with congenital anomalies, and maternal morbidities. The CDC includes the following conditions as congenital anomalies: anencephaly, meningomyelocele/spina bifida, cyanotic congenital heart disease, congenital diaphragmatic hernia, omphalocele, gastroschisis, limb reduction defect, cleft lip with or without cleft palate, cleft palate alone, down syndrome, suspected chromosomal disorder, and hypospadias. Maternal morbidity is defined by the CDC as the receipt or diagnosis of maternal transfusion, perineal laceration, ruptured uterus, unplanned hysterectomy, admission to intensive care unit.

Date cut-point based on Dobbs vs Jackson

The birth date cut-point was selected based on the assumption that pregnancy that was less than or at 22 weeks of gestation on June 24, 2022 resulted in early-to full-term birth (gestational age of 37 to 40 weeks) in October 2022. These are pregnancies that could have been theoretically affected by the decision. For the ease of interpretability of this article, births happening before and after October 2022 will be called, births pre/post-Dobbs ruling, respectively, to distinguish the births that could have been unaffected/affected by the ruling [17–21].

Statistical methods

The study periods were divided into pre-ruling (January 2021 – October 2022) and post-ruling (November 2022 – February 2024) birth phases. A series of CITS models was applied to assess the association of the *Roe v. Wade* overturning of maternal and infant health outcomes in states with and without abortion bans. The CITS method is a well-established quasi-experimental research design used to evaluate the impact of a policy or intervention in absence of opportunities for randomized control trials [15, 16, 19–21]. In this study, the CITS models compared post-Dobbs changes in health outcomes for each group (states with and without an abortion ban) with their own baseline trend. Thus, for any unobserved factor to cause a confounding issue, it would need to coincide with timing of Dobbs decision and cause a deviation

in ban states from their baseline trend while leaving the sample of unbanned states unaffected. Difference-in-differences (DID) method, another quasi-experimental research design, required the parallel trend assumption to ensure internal validity [19, 20]. To fulfill the parallel trend assumption, the difference between the banned and unbanned states had to be constant over time except for the impact of Dobbs decision [20]; however, our study data violated this assumption. Comparative interrupted time series did not rely on the fulfillment of this assumption, as the trend differences between the treatment and control groups were controlled [19].

More technically, the model specification included a group-specific linear time series to establish baseline trends in outcome rates (calculated per month) for both groups within the pre-ruling period (through the inclusion of “tau”, see Supplement Section B for the full model specification). Changes in trends specific to each group following the ruling were estimated and compared based on the legal status of abortion in each state. All models incorporated Newey–West corrected standard errors to adjust for seasonality in the data [21]. Ordinary least squares regressions were fitted using SAS, version 9.4, to evaluate the differences in outcomes. The study followed the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines for reporting cohort studies [22].

Sensitivity analyses

A series of sensitivity analyses were conducted with various cutoff dates. Outcomes of interest are associated with prematurity [23–26]. A birthdate cutoff of September 2022 was examined to include premature births resulting from pregnancies that were 22 weeks along on June 24, 2022, into the post-Dobbs ruling period. The 22-week gestational cutoff was chosen based on fetal viability and the likelihood of abortion restrictions affecting births. In most states, viability-based abortion limits are set around 22–24 weeks, meaning pregnancies reaching 22 weeks as of June 24, 2022 (Dobbs ruling) were likely impacted. This cutoff ensures that we capture births affected by abortion bans while excluding those that would have been carried to term regardless of policy changes. On the other hand, although abortion is permitted in states categorized as no-ban states, abortions at or after 21 weeks are uncommon (1% of all abortions) [27]. Early and full-term births resulting from pregnancies that were 16 to 20 weeks along June 24, 2022, would have occurred in November 2022. Thus, this alternate birth date cutoff was also examined.

Given the state-specific variations in defining abortion access restrictions, two additional analyses with varying classifications of states were conducted. These sensitivity

analyses include (1) the removal of Texas from the dataset and (2) the reclassification of states (see, appendix Table A5 for the lists of states) comparing strict “total ban” states, which enforced abortion bans under almost all circumstances, and “protected” states, which have not enacted or enforced a significant abortion restriction by the end of 2022, as outlined by Dench et al. [14].

Results

Baseline demographic characteristics are described in Table 1 by using 2021 birth records. Chi-square statistics showed that age at birth, race and ethnicity, education level, WIC participation status, rurality of residency, and birthplace were all statistically significantly different across the abortion ban status of states. Estimations of a series of CITS for non-living births, congenital anomalies, and maternal morbidities are summarized in Tables 2. Additional computations were performed to derive accessible numerical estimates from coefficients

presented in Table 2. These estimates (i.e. baseline mean, baseline trend) are presented in Table 3 to aid interpretability. Figures 1 and 2 illustrate the trend break in congenital anomalies and maternal morbidities outcomes before and after October 2022, highlighting the divergence in maternal and infant health outcomes between ban and non-ban states. These visualizations reinforce the statistical findings, demonstrating how outcomes shifted post-Dobbs.

Non-living births

For the outcome of non-living births, there were no statistically significant coefficient estimations, other than baseline means.

Congenital anomalies

At the baseline, the proportion of births with congenital anomalies was higher by 4.04 per 10 K births ($p = 0.002$; see, Table 2: *Treat*) among states with an abortion

Table 1 Maternal Demographic Characteristics by Abortion Ban Status (2021)

	States that enacted an abortion ban (14 states)	States that did not enact an abortion ban (25 states)	<i>p</i> -value ^c
Age at birth			
Less than 20	54,976 (5.8%)	53,882 (3.0%)	<.001
20–30	503,102 (52.8%)	712,586 (40.2%)	
30–40	369,649 (38.8%)	924,774 (52.2%)	
Above 40	25,325 (2.7%)	80,298 (4.5%)	
Race and ethnicity			
NH White	503,674 (52.8%)	868,290 (49.0%)	<.001
NH Black	150,502 (15.8%)	198,385 (11.2%)	
Hispanic	234,479 (24.6%)	463,336 (26.2%)	
Other	64,397 (6.8%)	241,529 (13.6%)	
Education			
Less than Highschool	122,427 (12.8%)	175,527 (9.9%)	<.001
High school/GED	281,824 (29.6%)	403,612 (22.8%)	
Some College	269,414 (28.3%)	447,966 (25.3%)	
Bachelor's and above	275,235 (28.9%)	698,039 (39.4%)	
WIC			
Yes	319,881 (33.6%)	490,188 (27.7%)	<.001
No	627,252 (65.8%)	1,254,667 (70.8%)	
Rurality ^a			
Metro	750,717 (78.8%)	1,621,115 (91.5%)	<.001
Nonmetro	202,335 (21.2%)	150,425 (8.5%)	
Birthplace			
Hospital/Clinic ^b	939,775 (98.6%)	1,743,018 (98.4%)	<.001
Home	12,215 (1.3%)	25,543 (1.4%)	
Other	1062 (0.1%)	2979 (0.2%)	
TOTAL	953,052	1,771,540	

^a Rurality is defined using the 2013 NCHS Urban–Rural Scheme for Counties

^b Hospital, Freestanding Birth Center, Clinics. Numbers provided in each category may not add up to the total due to missing values

^c Chi-Square *p*-value was calculated for categorical variables. Abbreviation: NH = Non-hispanic

Table 2 Estimates of the Association between the Dobbs decision and Non-Living Births, Births with Congenital Anomalies, and Births with Maternal Morbidities (Rates per 10 K Births; states with an abortion ban $N = 3,032,018$, without an abortion ban $N = 5,495,792$)

Variable	Coefficient Estimates
<i>Non-Living Births</i>	
τ	-0.07 (0.05)
<i>Post</i>	1.12 (0.66)
$\tau \times \text{Post}$	$> -0.01^a$ (0.08)
<i>Treat</i>	0.58 (1.18)
$\tau \times \text{Treat}$	0.06 (0.09)
<i>Treat</i> $\times$ <i>Post</i>	0.89 (1.25)
$\tau \times \text{Treat} \times \text{Post}$	-0.13 (0.12)
<i>Congenital Anomalies</i>	
τ	0.02 (0.04)
<i>Post</i>	-2.05 (0.55) ***
$\tau \times \text{Post}$	0.09 (0.05)
<i>Treat</i>	4.04 (1.24) **
$\tau \times \text{Treat}$	-0.34 (0.1) **
<i>Treat</i> $\times$ <i>Post</i>	4.31 (1.72) *
$\tau \times \text{Treat} \times \text{Post}$	0.42 (0.12) **
<i>Maternal Morbidities</i>	
τ	0.34 (0.07) ***
<i>Post</i>	4.78 (1.18) ***
$\tau \times \text{Post}$	0.11 (0.1)
<i>Treat</i>	-31.97 (0.89) ***
$\tau \times \text{Treat}$	-0.47 (0.08) ***
<i>Treat</i> $\times$ <i>Post</i>	-4.5 (1.79) *
$\tau \times \text{Treat} \times \text{Post}$	0.56 (0.15) ***

Regression estimates are from a comparative interrupted time series with Newey–West corrected standard errors to adjust for seasonality in the data. Standard errors are presented in parentheses. $^a > -0.01$ " indicates greater than -0.01 . Specifically, -0.004 was rounded

Not Living Births $R^2 = 0.2213$; Congenital Anomalies $R^2 = 0.3544$; Maternal Morbidities $R^2 = 0.9732$. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

τ (Tau): Pre-existing trend in each outcome prior to the Dobbs decision

Post: A binary variable indicating whether an observation occurred before (0) or after (1) Dobbs

Treat: A binary variable indicating whether a state is classified as a ban state (1) or a non-ban state (0)

Treat $\times$ *Post* (Interaction Term): Measures the differential effect of the Dobbs decision on ban vs. non-ban states by comparing pre- and post-Dobbs trends

$\tau \times \text{Post}$: Change in trend post-Dobbs in non-ban states

$\tau \times \text{Treat}$: Pre-existing trend difference between ban and non-ban states

$\tau \times \text{Treat} \times \text{Post}$: Difference in Changes in trend post-Dobbs in ban states compared to non-ban states

ban than those without a ban. In impacted births that occurred 22 months prior to the Dobbs decision, births with congenital anomalies generally decreased in states that would enact abortion bans. The magnitude of this decrease was estimated at an average of 0.33 per 10 K births ($p = 0.001$) in each month from January 2021

Table 3 Estimated Post-Dobbs Decision Birth Outcome Changes from the Baseline Trends (Rates per 10 K Births)

	States with an abortion ban	States without an abortion ban
<i>Non-Living Births</i>		
Baseline Mean	21.38 (1.36) ***	20.79 (0.70) ***
Baseline Trend	-0.01 (0.10)	-0.07 (0.05)
Post-Dobbs Change	2.02	1.12 (0.09)
Post-Dobbs Trend	-0.15 (0.17)	-0.07 (0.09)
<i>Congenital Anomalies</i>		
Baseline Mean	36.77 (1.36) ***	32.72 (0.58) ***
Baseline Trend	-0.33 (0.11) **	0.02 (0.04)
Post-Dobbs Change	2.27 (1.81)	-2.05 (0.55) ***
Post-Dobbs Trend	0.18 (0.17)	0.11 (0.06)
<i>Maternal Morbidities</i>		
Baseline Mean	117.46 (1.13) ***	149.42 (0.7) ***
Baseline Trend	-0.13 (0.1)	0.34 (0.07) ***
Post-Dobbs Change	0.28 (2.14)	4.78 (1.18) ***
Post-Dobbs Trend	0.53 (0.21) *	0.45 (0.12) ***

Calculations for the baseline trend and post-Dobbs decision change in the proportion of non-living births, births with congenital anomalies, and births with maternal morbidities for states without an abortion ban come directly from the regression estimates presented in Table 2. Post-Dobbs trend in states without an abortion ban is calculated as Post-Dobbs trend = $\tau + (\tau \times \text{Post})$

Calculations the proportion of non-living births, births with congenital anomalies, births with maternal morbidities for states with an abortion ban are as follows:

Baseline trend = $\tau + (\tau \times \text{Treat})$; post-Dobbs change = $\text{Post} + (\text{Treat} \times \text{Post})$; post-Dobbs trend = $\tau + (\tau \times \text{Treat}) + (\tau \times \text{Post}) + (\tau \times \text{Treat} \times \text{Post})$. Standard errors are presented in parentheses below the coefficient estimates.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

to October 2022 (see, *Baseline Trend* in Table 3). The monthly birth rates with congenital anomalies were generally unchanging for states without a ban in the same period.

Compared with the baseline trend, the number of births with congenital anomalies slightly increased (2.27 per 10 K; Table 3: *Post-Dobbs Change*) in the states with bans at the time of the Dobbs decision, but the increase was not statistically significant ($p = 0.213$). The states without bans saw a decrease in births with congenital anomalies by 2.05 per 10 K ($p < 0.001$; Table 3: *Post-Dobbs Change*) from the baseline. The change in the proportion of births with congenital anomalies was estimated to be higher at the rate of 4.31 per 10 K births ($p = 0.015$) among states with a ban compared to the states without such measure (see, Table 2: *Treat* $\times$ *Post*).

In terms of the change in trends, the rate of change in births with congenital anomalies went from the baseline decreasing trend to an increasing trend (0.18 per 10 K birth per month; Table 3: *Post-Dobbs Trend*) among states with an abortion ban. The trends remained steady among states that did not enact abortion bans during

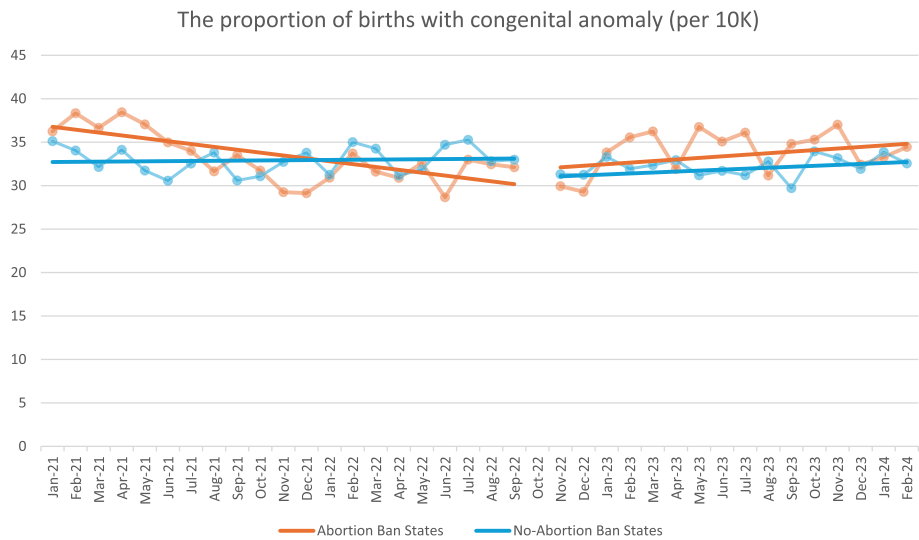


Fig. 1 Changes in the proportion of births with congenital anomalies (per 10 K) pre-/post-Dobbs decision by abortion ban status. This figure represents a crude proportion of births with congenital anomalies pre- and post-Dobbs by abortion ban status of states. Lines with markers represent crude proportions. Linear lines represent model derived trend lines. States with bans include Alabama, Arkansas, Idaho, Indiana, Kentucky, Louisiana, Mississippi, Missouri, North Dakota, Oklahoma, South Dakota, Tennessee, Texas, and West Virginia. States where abortion is legal beyond 22 weeks include California, Colorado, Connecticut, Delaware, the District of Columbia, Hawaii, Illinois, Maine, Maryland, Massachusetts, Michigan, Minnesota, Montana, Nevada, New Hampshire, New Jersey, New Mexico, New York, Oregon, Pennsylvania, Rhode Island, Vermont, Virginia, Washington, and Wyoming

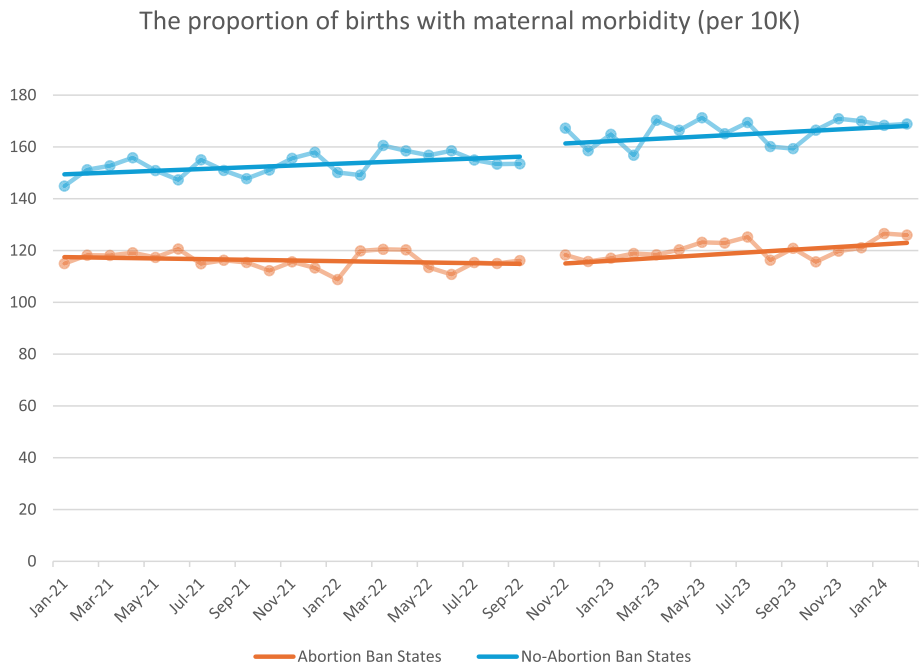


Fig. 2 Changes in the proportion of births with maternal morbidities (per 10 K) pre-/post-Dobbs decision by abortion ban status. This figure represents a crude proportion of births with maternal morbidities pre- and post-Dobbs by abortion ban status of states. Lines with markers represent crude proportions. Linear lines represent model derived trend lines. States with bans include Alabama, Arkansas, Idaho, Indiana, Kentucky, Louisiana, Mississippi, Missouri, North Dakota, Oklahoma, South Dakota, Tennessee, Texas, and West Virginia. States where abortion is legal beyond 22 weeks include California, Colorado, Connecticut, Delaware, the District of Columbia, Hawaii, Illinois, Maine, Maryland, Massachusetts, Michigan, Minnesota, Montana, Nevada, New Hampshire, New Jersey, New Mexico, New York, Oregon, Pennsylvania, Rhode Island, Vermont, Virginia, Washington, and Wyoming

the post-Dobbs period (*Baseline Trend* of 0.02 and *Post-Dobbs Trend* of 0.11 per 10 K per month; see, Table 3). Figure 1 presents trends in the proportion of births with congenital anomalies in states with universal abortion bans versus those without.

Maternal morbidities

The baseline proportion of births with maternal morbidities before the Dobbs decision was estimated to be 31.97 per 10 K ($p < 0.001$) lower among states that would enact an abortion ban than among counterpart states (see, Table 2: *Treat*). Prior to the Dobbs decision, the rate of births with maternal morbidity was unchanging among states that would enact an abortion ban (see, *Baseline Trend* in Table 3). Births with maternal morbidities increased at an average rate of 0.34 per 10 K ($p < 0.001$) per month leading up to October 2022 in states that did not enact an abortion ban (Table 3: *Baseline Trend*).

At the cutoff point distinguishing births that were potentially impacted by the Dobbs decision, the states with a ban did not see a deviation from the baseline trends in the proportion of births with maternal morbidities (see, Table 3: *Post-Dobbs Change*). On the other hand, an increase of 4.78 per 10 K births ($p < 0.001$) in the proportion of maternal morbidities was observed in states that did not enact such measures (Table 3: *Post-Dobbs Change*). The difference between the changes in the proportion of births with maternal morbidities across two groups of states was estimated to be 4.5 per 10 K ($p = 0.014$; Table 2: *Treat* $\times$ *Post*).

In states with an abortion ban, there was a change in the trends of the proportion of births with maternal morbidities across two period—as described above, there was no month-to-month change in the rate of births with maternal morbidities prior to the Dobbs decision; however, the rate was increasing by 0.53 per 10 K ($p = 0.011$) per month in the post-Dobbs period (Tables 3: *Post-Dobbs Trend*). The rate of change in the proportion of births with maternal morbidities remained unchanged among the states that did not enact abortion bans during the post-Dobbs period (Table 2: $\tau \times$ *Post*). Figure 2 presents trends in the proportion of births with maternal morbidities in states with universal abortion bans versus those without.

Sensitivity analyses

As described previously, a series of sensitivity analyses were conducted (2) without Newey–West corrected standard errors (2) with a birthdate cutoff of September 2022 (3) with a birthdate cut-point of November 2022 (4) with a removal of Texas from the dataset (5) with a classification of states [14] comparing strict “total ban” and “protected” states. These additional analyses produced

similar results for outcomes of non-living births (no differential immediate impact of the Dobbs decision across state policies) and congenital anomalies (a significantly more decrease in the rate of births with congenital anomalies among non-ban states compared to ban states immediately after the Dobbs decision). The finding of significant difference in the changes in the proportion of maternal morbidity pre- and post-Dobbs decision across states with and without abortion bans persisted when the analyses were performed without Newey–West corrected standard errors or with a removal of Texas. However, no significant difference in the immediate changes in the proportion of maternal morbidity due to the Dobbs decision across abortion ban policies was observed when the analyses were performed with alternative birthdate cutoffs of September and November 2022. See, Appendix Tables A1–A5 for the detailed results of the CITS estimation.

Discussion

This study provides one of the initial insights into the public health implications of the Dobbs v. Jackson Women’s Health Organization decision. Our findings reveal that states with abortion bans had a higher rate of births with congenital anomalies prior to the Dobbs decision. Although states with abortion bans saw an increase in rates of births with congenital anomalies across pre- and post-Dobbs decision, the increase was not statistically significant. However, the states that did not enact abortion ban laws had significant decreases in the rates of births with congenital anomalies, resulting in a significant exacerbation of the disparity of the rates of births with congenital anomalies post-Dobbs across states with and without such abortion restrictions. While ban states saw restricted access and healthcare options in the cases of pregnancy with known congenital anomalies, non-ban states may have indirectly benefited from the nationwide dialogues and shifts in pregnancy planning, prenatal care, and abortion-related healthcare utilization. Conversely, the records of maternal morbidity increased significantly more in states without abortion bans compared to states with abortion bans. These results highlight the complexity of assessing the observed differences of the abortion legislation on public health.

Declercq et al. found that states with abortion bans or restrictions had higher neonatal death rates in the first 27 days of life, as well as higher post neonatal mortality rates between 28 and 365 days after birth [7]. The limitation in medical decision-making due to inability of parents and providers to terminate a pregnancy when a potentially fatal or life-limiting anomaly is detected, such as anencephaly, is likely to increase such births in a long term. Additionally, abortion-restricted states had fewer

maternal care providers, including a 32% lower ratio of obstetricians to births [28]. This further complicates access to birthing services and prenatal care, potentially leading to the difference in the rates of birth with congenital anomalies observed in our study.

Infants with congenital anomalies have special medical needs. Supporting families of these infants, who are at risk of financial, emotional, physical and mental health, and logistical and parenting challenges, is critical [9–11]. Historically, many states in which abortion is restricted have low funding to support families with children with special health needs [29]. Our results suggest that the health disparity may increase without expanded resources to support these infants and families in need. It is essential to allocate the appropriate resources for long-term support to these families. This underscores the necessity of examining congenital anomalies as a critical outcome when assessing the role of the *Dobbs* ruling.

Maternal morbidity was observed to increase at higher rates immediately after the *Dobbs* decision in states without abortion bans in the primary analysis as well as two sensitivity analyses, although this finding did not hold in three additional analyses and was sensitive to the timing of birthdate cut-point and state classification (ban/no ban). In every analysis, the overall maternal morbidity rate was higher in the states that did not impose abortion bans compared to those that did. The finding suggesting maternal morbidity increased at higher rates in states without abortion bans after the *Dobbs* decision may seem counterintuitive, given that the abortion restriction has been linked with higher maternal mortality [7]. The severity of maternal morbidity and mortality metrics, while related, do not necessarily align, with risk factors affecting each outcome differently. The overall higher maternal morbidity rates in the states without an abortion ban may be due to women in these states having better access to healthcare services, leading to more extensive reporting and management of maternal morbidity. The measurement of maternal morbidity has been documented to be dependent on the reporting practice and coding systems, with inconsistent coding across states [30, 31]. One potential explanation for the observed increase in maternal morbidity immediately after the *Dobbs* decision in the states that did not enact abortion ban laws may be due to the national tension leading to more conscious or over-reporting of maternal morbidity.

It is important to acknowledge the nuances in classifying states as ban or non-ban due to variations in enforcement timelines, pre-existing restrictions, and legal ambiguity. While most abortion bans took effect shortly after the *Dobbs* decision, some states faced delays or legal uncertainty, affecting access differently. Indiana

and North Dakota were classified as ban states, despite implementing their bans later. Texas had already imposed severe abortion restrictions via SB8 in September 2021, effectively functioning as a ban state prior to *Dobbs* and making its policy landscape distinct. Wisconsin, while lacking an official post-*Dobbs* ban, reverted to an 1849 law, leading abortion providers to suspend services for months due to legal uncertainty. Idaho and West Virginia's bans aligned more closely with the initial wave of post-*Dobbs* abortion bans. Pennsylvania and Wyoming add further complexity as non-ban states. Pennsylvania has maintained legal abortion access, but state leadership remains divided, creating policy uncertainty. Wyoming has passed an abortion ban that has not yet taken effect, signaling hostility toward abortion even though access remains legally protected for now. These state-specific variations underscore the challenges in defining abortion access restrictions and examining their potential impact on maternal and fetal health outcomes.

Further research should investigate racial implications, given that increased restriction in accessing abortion creates an even more dangerous climate for Black women, who are 2–4 times as likely to experience maternal mortality and morbidity than their White counterparts [32]. Moreover, many laws restricting abortion were purported to protect women's health by regulating abortion facilities as ambulatory surgery centers, dictating the types of healthcare professionals who can perform abortions, and requiring admitting privileges to hospitals for clinicians who provide abortion care [33].

In response to these ongoing challenges, the executive order issued in 2024 to advance women's health research and innovation aims to enhance the quality and accessibility of healthcare for women and improve health outcomes through coordinated efforts [34]. This order highlights the importance of investing in women's health and the need for comprehensive policies to support maternal and infant health in the wake of the *Dobbs* decision.

Strengths and limitations

Our study has several strengths. First, it uses a robust CITS design to assess the association of the *Dobbs* decision on maternal and fetal outcomes, allowing for a clear comparison of trends in states with and without abortion bans. Second, the use of a national dataset (CDC WONDER) ensures that our findings are based on high-quality, representative data. Finally, a series of sensitivity analyses was conducted to augment the main analyses.

However, there are some limitations to this study. The timeframe post-overturning of *Roe v. Wade* may be too short to measure the full effect. The effect may have been ongoing even before the formal overturning due

to already restricted access. Additionally, we were only able to capture where births occurred, and not the state of residence. Women may have traveled to give birth, which could have introduced residual confounding factors. The CDC-WONDER database does not include information on the date of conception. As such, we were not able to identify pregnancies less than or at 22 weeks of gestation at the time of the Dobbs decision. The birth date cutoff was instead based on the assumption that most births were carried to early or full term. Prematurity is associated with all outcomes in this study [23–26]. A sensitivity analysis that includes premature births that were 22 weeks along into the post-Dobbs decision resulted in similar results. While the CITS design enables separating the policy impact from existing secular trends, unobserved factors—such as social, economic, and demographic differences—may have contributed to the observed trends, particularly given the ongoing COVID-19 pandemic during this period. Additionally, the included maternal morbidity outcomes vary in severity, which may influence the interpretation of overall trends in maternal health.

Conclusions

Our examination did not find direct evidence of abortion bans increasing adverse birth outcomes. However, the difference in the changes in congenital anomalies suggests that there is a potential for the exacerbation of inequities [32], and further investigation is warranted. Increasing federal funding for reproductive healthcare, family planning, and maternity care could help mitigate the impact of state abortion bans and the potential further fragmentation of the maternal health system. Educating policymakers to protect fetal and maternal health is crucial. Actions are needed by all stakeholders to safeguard the health and to ensure optimal and equitable outcomes for mothers and infants through coordinated efforts.

Abbreviations

CDC WONDER	The Centers for Disease Control and Prevention's Wide-ranging ONline Data for Epidemiologic Research
U.S.	United States
Dobbs decision	The Supreme Court decision in Dobbs v. Jackson Women's Health Organization
CITS	Comparative interrupted time series

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

The study design and concept were conceived by LG and KL. GL conducted the data curation, and KL conducted the statistical analysis plan. Both authors prepared the first draft of the manuscript and provided edits. Both authors read and approved the final manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are available in the the Centers for Disease Control and Prevention's Wide-ranging ONline Data for Epidemiologic Research (CDC-WONDER), Natality Information, <https://wonder.cdc.gov/natality.html>. The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was reviewed and approved by the University of Arkansas for Medical Sciences Institutional Review Board (#298211).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Guttacher Institute. Interactive Map: US Abortion Policies and Access After Roe. May 15, 2024. <https://states.guttacher.org/policies/>. Accessed May 20, 2024.
- Cai W, Johnston T, McCann A, Schoenfeld Walker A. Half of US women risk losing abortion access without Roe. The New York Times. 2022. <https://www.nytimes.com/interactive/2022/05/07/us/abortion-access-ro-v-wade.html>. Accessed 20 May 2024.
- Vricella LK, Gawron LM, Louis JM. Society for Maternal-Fetal Medicine (SMFM) Consult Series #48: Immediate postpartum long-acting reversible contraception for women at high risk for medical complications. Am J Obstet Gynecol. 2019;220(5):B2–12. <https://doi.org/10.1016/j.ajog.2019.02.011>.
- Pabayo R, Ehntholt A, Cook DM, Reynolds M, Muennig P, Liu SY. Laws restricting access to abortion services and infant mortality risk in the United States. Int J Environ Res Public Health. 2020;17(11):3773. <https://doi.org/10.3390/ijerph17113773>.
- Gemmill A, Franks AM, Anjur-Dietrich S, et al. US abortion bans and infant mortality. JAMA. 2025. <https://doi.org/10.1001/jama.2024.28517>.
- Haddad LB, Nour NM. Unsafe abortion: unnecessary maternal mortality. Rev Obstet Gynecol. 2009;2(2):122–6.
- Declercq E, Barnard-Mayers R, Zephyrin LC, Johnson K. The U.S. Maternal Health Divide: The Limited Maternal Health Services and Worse Outcomes of States Proposing New Abortion Restrictions. Commonwealth Fund. 2022. <https://www.commonwealthfund.org/publications/issue-briefs/2022/dec/us-maternal-health-divide-limited-services-worse-outcomes>. Accessed 20 May 2024.

8. Miller HE, Fraz F, Zhang J, et al. Abortion Bans and Resource Utilization for Congenital Heart Disease: A Decision Analysis. *Obstet Gynecol*. 2023;142(3):652–9. <https://doi.org/10.1097/AOG.0000000000005291>.
9. McClung N, Glidewell J, Farr SL. Financial burdens and mental health needs in families of children with congenital heart disease. *Congenit Heart Dis*. 2018;13(4):554–62. <https://doi.org/10.1111/chd.12605>.
10. Shah NR, Kim KM, Wong V, et al. Mothers of children with major congenital anomalies have increased health care utilization over a 20-year post-birth time horizon. *PLoS ONE*. 2021;16(12): e0260962. <https://doi.org/10.1371/journal.pone.0260962>.
11. Dabrowska A, Pisula E. Parenting stress and coping styles in mothers and fathers of pre-school children with autism and Down syndrome. *J Intellect Disabil Res*. 2010;54(3):266–80. <https://doi.org/10.1111/j.1365-2788.2010.01258.x>.
12. Polic A, Rapkin RB. Access to abortion training. *Semin Perinatol*. 2020;44(5): 151271. <https://doi.org/10.1016/j.semperi.2020.151271>.
13. Caraher R. Do Abortion Bans Affect Reproductive and Infant Health? Evidence from Texas's 2021 Ban and Its Impact on Health Disparities. 2024. <https://doi.org/10.2139/ssrn.4911886>.
14. Dench D, Pineda-Torres M, Myers CK. The Effects of the Dobbs Decision on Fertility. *SSRN Electronic J*. 2023. <https://doi.org/10.2139/ssrn.4636864>. Published Online.
15. Bernal JL, Cummins S, Gasparrini A. The use of controls in interrupted time series studies of public health interventions. *Int J Epidemiol*. 2018. <https://doi.org/10.1093/ije/dyy135>.
16. Linden A. Conducting interrupted time-series analysis for single- and multiple-group comparisons. *Stata J*. 2015. <https://doi.org/10.1177/1536867x1501500208>.
17. Centers for Disease Control and Prevention. Natality Information - Live Births. Published online October 3, 2023. <https://wonder.cdc.gov/natality.html>. Accessed 20 May 2024
18. Kaiser Family Foundation. Abortion Policy Tracker. November 2023. <https://www.kff.org/other/state-indicator/abortion-policy-tracker/?currentTimeframe=0&sortModel=%7B%22colId%22:%22Location%22,%22sort%22:%22asc%22%7D>. Accessed 20 May 2024.
19. Somers MA, Zhu P, Jacob R, Bloom H. The Validity and Precision of the Comparative Interrupted Time Series Design and the Difference-in-Difference Design in Educational Evaluation. *Mdrc*. 2012:27–45.
20. Imbens GW, Wooldridge JM. What's New in Econometrics? Lecture 10 Difference-in-Differences Estimation. 2007. https://www.nber.org/sites/default/files/2021-03/slides_10_diffindiffs.pdf
21. Linden A. Conducting Interrupted Time-series Analysis for Single- and Multiple-group Comparisons. *Stata J Promoting Commun Stat Stata*. 2015;15(2):480–500. <https://doi.org/10.1177/1536867X1501500208>.
22. Cuschieri S. The STROBE guidelines. *Saudi J Anaesth*. 2019;13(Suppl 1):S31–4. https://doi.org/10.4103/sja.SJA_543_18.
23. Gunn-Charlton JK. Impact of comorbid prematurity and congenital anomalies: a review. *Front Physiol*. 2022. <https://doi.org/10.3389/fphys.2022.880891>.
24. Kase JS, Visintainer P. The relationship between congenital malformations and preterm birth. *J Perinat Med*. 2007;35(6):538–42. <https://doi.org/10.1515/JPM.2007.132>.
25. Phillips JM, Bodnar LM, Himes KP. Association between gestational age at delivery and indicator-specific severe maternal morbidity. *J Matern Fetal Neonatal Med*. 2023;36(1): 2198633. <https://doi.org/10.1080/14767058.2023.2198633>.
26. Centers for Disease Control and Prevention. Preterm Birth. Maternal Infant Health. <https://www.cdc.gov/maternal-infant-health/preterm-birth/index.html>
27. Gomez I, Salganicoff A, Sobel L. Abortions Later in Pregnancy in a Post-Dobbs Era. Kaiser Family Foundation. 2024. <https://www.kff.org/womens-health-policy/issue-brief/abortions-later-in-pregnancy-in-a-post-dobbs-era/>. Accessed 3 June 2024.
28. American Society for Reproductive Medicine. States' Abortion Laws: Potential Implications for Reproductive Medicine. 2022. https://www.asrm.org/globalassets/_asrm/advocacy-and-policy/dobbs/state_abortion_laws_p2_oct_22.pdf. Accessed 20 May 2024.
29. Williams E, Musumeci M. Children with Special Health Care Needs: Coverage, Affordability, and HCBS Access. Kaiser Family Foundation. <https://www.kff.org/medicaid/issue-brief/children-with-special-health-care-needs-coverage-affordability-and-hcbs-access/>. Accessed 5 June 2024.
30. Hirai AH, Owens PL, Reid LD, Vladutiu CJ, Main EK. Trends in severe maternal morbidity in the US across the transition to ICD-10-CM/PCS from 2012–2019. *JAMA Netw Open*. 2022. <https://doi.org/10.1001/jamanetworkopen.2022.22966>.
31. Hirai AH, Owens PL, Reid LD, Vladutiu CJ, Main EK. Associations between state-level severe maternal morbidity and other perinatal indicators. *JAMA Netw Open*. 2022. <https://doi.org/10.1001/jamanetworkopen.2022.24621>.
32. Creanga AA, Bateman BT, Kuklina EV, Callaghan WM. Racial and ethnic disparities in severe maternal morbidity: a multistate analysis, 2008–2010. *Am J Obstet Gynecol*. 2014;210(5):435.e1–435.e8. <https://doi.org/10.1016/j.ajog.2013.11.039>.
33. Guttmacher Institute. State laws and policies: Targeted Regulation of Abortion Providers. 2023. <https://www.guttmacher.org/state-policy/explore/targeted-regulation-abortion-providers>. Accessed 10 Sept 2024.
34. Biden JR. Executive Order on Advancing Women's Health Research and Innovation. The White House. 2024. <https://www.whitehouse.gov/briefing-room/presidential-actions/2024/03/18/executive-order-on-advancing-womens-health-research-and-innovation/>. Accessed 16 June 2024.

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